

Adjuvant Ab Interno Tumor Treatment After Proton Beam Irradiation



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• **PURPOSE:** This study was performed to show long-term outcomes concerning globe preservation in uveal melanoma patients after proton beam therapy with the main focus on outcomes according to different adjuvant ab interno surgical procedures.

• **DESIGN:** Retrospective cohort study.

• **METHODS:** All patients treated with primary proton beam therapy for choroidal or ciliary body melanoma between June 1998 and June 2015 were included.

• **RESULTS:** A total of 2499 patients underwent primary proton beam therapy, with local tumor control and globe preservation rates of 95.9% and 94.8% after 5 years, respectively. A total of 110 (4.4%) patients required secondary enucleation. Unresponsive neovascular glaucoma was the leading cause of secondary enucleation in 78 of the 2499 patients (3.1%). The 5-year enucleation-free survival rate was 94.8% in the endoresection group, 94.3% in the endodrainage group, and 93.5% in the comparator group. The log-rank test showed $P = .014$ (comparator group vs endoresection group) and $P = .06$ (comparator group vs endodrainage-vitreectomy group). Patients treated with endoresection or endodrainage-vitreectomy developed less radiation retinopathy (30.5% and 37.4% after 5 years, $P = .001$ and $P = .048$ [Kaplan-Meier], respectively) and less neovascular glaucoma (11.6% and 21.3% after 5 years, $P = .001$ and $P = .01$ [Kaplan-Meier], respectively) compared with the comparator group (52.3% radiation retinopathy and 57.8% neovascular glaucoma after 5 years).

• **CONCLUSION:** This study suggests that in larger tumors the enucleation and neovascular glaucoma rates might be reduced by adjuvant surgical procedures. Although endoresection is the most promising adjuvant treatment option, the endodrainage-vitreectomy is recommended in patients who are ineligible for endoresection. (*Am J Ophthalmol* 2017;178:94–100.

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IT IS KNOWN THAT PROTON BEAM THERAPY, PARTICULARLY in larger tumors, can lead to inflammation and neovascular glaucoma (“toxic tumor syndrome”) by leaving an ischemic or exudative tumor scar after irradiation.^{1–3} For a very long time, patients were only observed after irradiation because the exudative component of the toxic tumor, the exudative retinal detachment, which often even enlarges after proton beam therapy, reattaches by itself. Therefore, some years ago many patients did not receive any adjuvant surgical treatment after proton beam therapy. In 1998, Damato and associates first described the endoresection procedure in which the tumor was resected by vitrectomy.⁴ This procedure was gradually being used in our daily clinical routine for tumors exceeding 6–7 mm in thickness with an accompanying exudative retinal detachment. Because of an elevated risk of perioperative hemorrhage in some patients who presented with characteristics in terms of large tumor diameter, nonreplaceable anticoagulants, or severe hypertension, endoresection was not a viable treatment option. Therefore, the procedure first described by Gibran and Kapoor in 2009, in which the exudative retinal detachment was reattached by vitrectomy after a primary irradiation, was performed in those patients who were ineligible for endoresection.⁵ We have named this procedure endodrainage-vitreectomy, and a first series report of patients who underwent the endodrainage-vitreectomy in our department was published in 2014.⁶ Thus, 3 main options have been used after primary proton beam therapy: endoresection, endodrainage-vitreectomy, and observation. The aim of this study is to assess the secondary enucleation rate with respect to adjuvant surgical treatment options that can be performed by pars plana vitrectomy after primary proton beam therapy. Further, we report on different reasons leading to secondary enucleation and complications threatening globe preservation.

METHODS

THIS STUDY WAS APPROVED BY THE INSTITUTIONAL REVIEW board of the Charité-Universitätsmedizin Berlin, Berlin, Germany, and was in accordance with the tenets of the Declaration of Helsinki. All patients with diagnosed choroidal or ciliary body melanoma that were treated with primary proton beam therapy between June 1998 and June 2015 were included. In our oncology service, proton beam therapy is applied to centrally located tumors close to

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